SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE:CSA1741

CO1-AT3-Descriptive Test

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COURSE OUTCOME COVERED

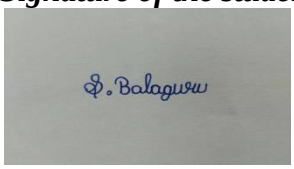
CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weight age: 10%

Code of Conduct:

I Balaguru S (192312070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1. **A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.**

1. Understanding of Concepts (5 Marks)

An intelligent agent is a system that perceives its environment through sensors and performs actions to achieve desired outcomes. In a smart home automation system, sensors collect information such as motion, temperature, light intensity, and time of day. The agent processes this information and controls devices such as lights, air conditioners, thermostats, and security alarms.

The four major types of intelligent agents are:

1. Simple Reflex Agent
2. Model-Based Reflex Agent
3. Goal-Based Agent
4. Utility-Based Agent

2. Explanation of Different Intelligent Agents (5 Marks)

A. Simple Reflex Agent

A simple reflex agent selects actions based only on the current percept using predefined condition-action rules. It does not maintain a history of past observations or learn user preferences.

Working in a Smart Home:

- If motion is detected in a room, switch ON the lights.
- If the temperature exceeds 28°C, switch ON the air conditioner.
- If an unauthorized entry is detected, activate the alarm.

Advantages:

- Simple to design and implement.
- Responds quickly to real-time inputs.
- Requires fewer computational resources.

Limitations:

- Cannot remember previous states.
- Cannot learn user behaviour.
- May waste energy by repeatedly activating devices without considering context.

Example: A motion sensor switches ON the hallway light whenever movement is detected, regardless of whether it is daytime or nighttime.

B. Model-Based Reflex Agent

A model-based reflex agent maintains an internal state representing information about the environment. It uses current sensor inputs and previous observations to update its understanding of the home.

Working in a Smart Home:

- Remembers whether a room is occupied.
- Tracks the current temperature and the operating state of the air conditioner.
- Uses door sensor information to determine whether an entry event is unexpected.
- Considers ambient light levels before switching ON lights.

Advantages:

- Handles partially observable environments.
- Uses previous information to improve decisions.
- Avoids unnecessary device activation by considering the current state.

Limitations:

- Requires an accurate internal model.
- May make incorrect decisions if sensor data or stored information is outdated.
- Does not automatically learn preferences unless a learning mechanism is added.

Example: If the room is already illuminated and occupied, the agent avoids switching ON additional lights unnecessarily.

C. Goal-Based Agent

A goal-based agent selects actions by evaluating whether they help achieve a specified goal. It may compare possible actions and plan a sequence of actions to reach the desired state.

Working in a Smart Home:

- Goal: Maintain a comfortable indoor temperature.
- Goal: Ensure the home is secure when occupants leave.
- Goal: Reduce electricity consumption when rooms are unoccupied.

For example, when the user leaves home, the agent may switch OFF unnecessary lights, adjust the thermostat, and activate the security system.

Advantages:

- Supports planning and decision-making.
- Can choose different actions to achieve the same goal.
- Adapts its action sequence to changing conditions.

Limitations:

- May not determine the best trade-off between competing goals.
- Planning can require additional computation.
- Requires clearly defined goals.

Example: To achieve the goal of securing an empty house, the agent locks smart doors, activates security sensors, and switches OFF unnecessary appliances.

D. Utility-Based Agent

A utility-based agent evaluates possible actions using a utility function that represents the desirability of their outcomes. It selects the action with the highest expected utility while considering competing objectives.

Working in a Smart Home:

The agent balances:

- Comfort: Maintaining a preferred temperature.
- Energy efficiency: Minimizing electricity consumption.
- Security: Detecting and responding to threats.
- Convenience: Learning and adapting to user preferences.

For example, instead of running the air conditioner continuously, the agent may choose a slightly higher temperature when the room is empty to save energy while maintaining acceptable comfort.

Advantages:

- Balances multiple competing objectives.
- Supports cost-effective decisions.
- Can evaluate uncertain outcomes using expected utility.
- Can incorporate learned user preferences.

Limitations:

- Designing an appropriate utility function is difficult.
- Incorrect weights may prioritize energy savings over comfort or security.
- Learning user preferences requires suitable data and a learning mechanism.

Example: When the user leaves a room, the agent switches OFF the lights immediately but adjusts the air conditioner according to the room temperature, expected occupancy, and energy-saving policy.

3. Application and Comparative Analysis (4 Marks)

Agent Type	Smart Home Function	Main Advantage	Main Limitation
Simple Reflex	Switches lights ON when motion is detected.	Fast response	Ignores history and context
Model-Based	Tracks occupancy, temperature, and device states.	Maintains environmental state	Depends on the accuracy of its internal model
Goal-Based	Plans actions to achieve comfort, energy-saving, and security goals.	Supports planning	Does not inherently compare trade-offs
Utility-Based	Balances comfort, energy cost, and security.	Optimizes competing objectives	Requires a well-designed utility function

Example Scenario

Suppose the user leaves home at 9:00 AM while the air conditioner and lights are ON.

- **Simple Reflex Agent:** Switches OFF lights if a suitable absence-related rule is triggered by a sensor.
- **Model-Based Agent:** Uses occupancy history and device states to determine that the house is likely unoccupied.
- **Goal-Based Agent:** Plans to reduce energy consumption and activate the security system.
- **Utility-Based Agent:** Selects actions that balance energy savings, security, and the user's expected return time.

The utility-based approach provides the most flexible decision-making because it can evaluate the relative benefits and costs of different actions.

4. Organization and Recommended Hybrid Approach (3 Marks)

The most effective design is a **hybrid intelligent agent combining simple reflex, model-based, goal-based, and utility-based mechanisms**.

The system can be organized as follows:

1. **Simple Reflex Layer:** Handles immediate safety-critical events, such as triggering an alarm when a verified intrusion is detected.
2. **Model-Based Layer:** Maintains the current state of rooms, occupancy, temperature, lighting, and security devices.
3. **Goal-Based Layer:** Establishes objectives such as maintaining comfort, securing the home, and reducing electricity use.
4. **Utility-Based Layer:** Evaluates alternative actions and selects the best trade-off among comfort, energy efficiency, security, and convenience.
5. **Learning Layer:** Analyzes user behaviour over time to update preferences, schedules, and comfort settings.

The learning layer can use historical temperature preferences, lighting schedules, and occupancy patterns. Learned preferences should be updated carefully and should not override essential security rules.

5. Accuracy and Justification (2 Marks)

A hybrid approach is the most suitable because no single agent type handles every requirement effectively. Simple reflex rules provide fast responses, model-based reasoning handles incomplete

environmental information, goal-based planning achieves specified objectives, and utility-based reasoning balances competing priorities.

Security-critical actions should have higher priority than energy savings or convenience. The system should also include manual overrides, privacy protection, and safe fallback behaviour when sensors fail.

Result:

A hybrid intelligent agent with a utility-based decision-making mechanism and a learning layer provides an effective smart home automation solution. It enables real-time responses, adapts to user preferences, improves energy efficiency, maintains comfort, and prioritizes security.

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2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

1. Understanding of Concepts (5 Marks)

A robot placed in an unknown maze must find a path from its starting position to the exit without knowing the complete layout in advance. It must explore the environment, identify obstacles, record visited locations, and select actions that lead toward the goal.

Uninformed search algorithms solve problems without using heuristic information about the distance to the goal. Two suitable strategies are:

1. **Uniform Cost Search (UCS):** Expands the node with the lowest accumulated path cost.
2. **Iterative Deepening Search (IDS):** Repeatedly performs depth-limited search with increasing depth limits until the goal is found.

The robot can also be designed using different intelligent agent types, such as simple reflex, model-based, goal-based, and utility-based agents.

2. Explanation of UCS and IDS (5 Marks)

A. Uniform Cost Search (UCS)

UCS selects the node with the smallest total cost from the starting position. It uses a priority queue to maintain the order of exploration.

Working Steps:

1. Place the starting node in the priority queue with cost 0.
2. Remove the node with the lowest path cost.
3. Check whether the node is the exit.
4. If it is not the exit, identify its reachable neighboring cells.
5. Calculate the new path cost for each neighbor.
6. Update the priority queue when a lower-cost path is discovered.
7. Repeat until the exit is reached or no reachable nodes remain.

Advantages:

- Finds an optimal path when movement costs are non-negative.
- Handles different movement costs.
- Systematically explores paths according to their accumulated cost.

Disadvantages:

- May explore many unnecessary nodes.
- Can require large amounts of memory to store the priority queue and explored nodes.
- Does not use information about the goal's direction.

B. Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited depth-first search, increasing the depth limit after each unsuccessful iteration.

Working Steps:

1. Set the initial depth limit to 0.
2. Search the maze up to the current depth limit.
3. If the exit is found, return the path.
4. Otherwise, increase the depth limit by 1.
5. Restart the search from the starting position.
6. Continue until the exit is found or the search is exhausted.

Advantages:

- Requires less memory than UCS and ordinary breadth-first search.
- Complete for finite branching factors when the goal is reachable.
- Finds a shortest path when all movements have equal cost.

Disadvantages:

- Repeats the exploration of nodes at shallow depths.
- May be inefficient for deep solutions.
- Does not guarantee the minimum-cost path when movement costs differ.

3. Application and Time-Space Complexity (4 Marks)

Let:

- (b) = branching factor, or average number of successors per node.
- (d) = depth of the shallowest goal.
- (C^*) = optimal solution cost.
- (\varepsilon) = minimum positive step cost.

Feature	Uniform Cost Search	Iterative Deepening Search
Node selection	Lowest accumulated path cost	Depth limit increases gradually
Time complexity	Commonly expressed as (O(b^{1+\lfloor C^*/\varepsilon\rfloor})) under standard assumptions	(O(b^d))
Space complexity	Exponential in the worst case; stores frontier and explored nodes	(O(bd)) for depth-first search stack and path
Completeness	Complete with positive step costs and finite branching	Complete with finite branching
Optimality	Optimal for non-negative costs under standard UCS conditions	Optimal when all actions have equal cost
Main use	Unequal movement costs	Equal movement costs and limited memory

Example:

Suppose the robot has two possible routes to the exit:

- Route A: 5 movements, each costing 1; total cost = 5.

- Route B: 3 movements, each costing 2; total cost = 6.

UCS selects Route A because its total cost is lower. IDS may find Route B first because it has a smaller depth, even though it is more expensive. Therefore, UCS is more appropriate when minimizing total cost is essential.

The stated complexities are theoretical bounds. Actual performance depends on the maze structure, branching factor, movement costs, and duplicate-state handling.

4. Intelligent Agents and Their Effect on Decision-Making (3 Marks)

An intelligent agent perceives its environment through sensors and selects actions to achieve its objectives.

Agent Type	Function in the Maze	Effect on Decision-Making
Simple Reflex Agent	Chooses actions using current sensor information and predefined rules.	Responds quickly but may revisit locations or get trapped in loops.
Model-Based Agent	Maintains an internal map of discovered paths, walls, and visited cells.	Makes better decisions using previous observations.
Goal-Based Agent	Selects actions that help reach the exit.	Supports planning toward the destination.
Utility-Based Agent	Evaluates alternatives based on cost, risk, and other preferences.	Balances safety, efficiency, and movement cost.

For an unknown maze, a model-based agent is especially useful because the robot cannot observe the entire environment at once. It can update its internal map as new cells are explored.

A goal-based component helps it focus on reaching the exit, while a utility-based component can be added if the robot must balance travel time, battery usage, and risk.

5. Recommended Combination and Justification (2 Marks)

The most effective design is a **model-based, goal-directed agent using UCS with online exploration**.

1. The robot initially explores the maze using its available sensors.
2. It records discovered walls, open paths, and visited locations in an internal map.
3. UCS selects the lowest-cost route among the currently known reachable paths.
4. When the robot discovers new obstacles or passages, it updates the map and replans.
5. The robot continues until it reaches the exit.

Justification:

- The model-based agent handles incomplete knowledge by maintaining an updated internal representation.
- UCS minimizes total path cost when movement costs are non-negative.
- Online exploration allows the robot to adapt as new information becomes available.
- A visited-state record helps prevent unnecessary repeated exploration.

IDS is a good alternative when all moves have equal cost and memory is very limited. However, UCS is preferable when the cost of different paths varies.

Result: A model-based, goal-directed robot using UCS with online exploration is the most suitable choice when the objective is to find a minimum-cost path through an initially unknown maze. The robot must update its map and replan whenever new information changes the available routes

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